Jiatong Ma

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Portfolio: jt-portfolio-snowy.vercel.app

EDUCATION

Beijing University of Posts and Telecommunications

Joint Programme with Queen Mary University of London *Sep 2023 – Jul 2027*

- Bachelor of Engineering in Digital Media Technology, International School
- GPA: 3.20/4.00
- Selected Coursework: Design & Build Project I/II (95/95), Interaction Design Foundations (94), Fundamentals of Web Technology (91), Mathematical Foundations of Data Science (88), Creation of Digital Animation (88), Sound Recording and Production Techniques (87), Digital Photography and Video Post-production (85), Technical English Language and Study Skills I (85)

PUBLICATIONS

- Chen, Peng; **Ma, Jiatong**; Zhong, Liqiao. "Extending Emotional Well-Being in Digital Art Therapy: The Role of Dynamic Audiovisual Feedback." *Proceedings of CONF-SPML 2026 Symposium: The 2nd Neural Computing and Applications Workshop 2025*, pp. 152–168, published Nov. 26, 2025. DOI: 10.54254/2755-2721/2026.TJ30075.

RESEARCH EXPERIENCE

Dynamic Audiovisual Feedback for Digital Art Therapy

Jul – Nov 2025

Co-First Author, advised by Lorie Loeb, Research Professor, Department of Computer Science, Dartmouth College

- Examined the capacity of dynamic audiovisual feedback in a keyboard-controlled digital art installation to support short-term emotional regulation among 10 urban students aged 17–21.
- Developed the experimental system by mapping keyboard inputs to natural visual transitions and therapeutic audio feedback, thereby constructing a closed input-feedback-emotion interaction loop.
- Contributed to the methodological design and academic writing by specifying three within-subject conditions, PANAS-based affect measurement, interaction-count logging, and total-duration tracking.
- Processed and analyzed affective-scale and interaction-behavior data with Python; multimodal feedback significantly reduced Distressed scores and increased Attentive scores, with engagement further reflected in higher interaction frequency.

INTERNSHIP EXPERIENCE

MiraclePlus

Jan – Mar 2026

AI Sector Research Intern, Research Department

- Systematically mapped frontier AI ecosystems across LLMs, RAG, agentic AI, MLSys, AI infrastructure, and world-model research, constructing a 200+ researcher database annotated by institution, publication, research direction, and commercial relevance.
- Independently conducted 24 interviews with AI researchers, including paper authors and research-institute affiliates, and synthesized technical decisions, commercialization constraints, and venture-relevance signals into structured research notes.
- Triangulated publications, researcher profiles, and interview findings to evaluate product opportunities in agentic AI, AI infrastructure, and world models, supporting internal discussions on early-stage venture directions.
- Developed and deployed an OpenClaw-based workflow for researcher discovery, homepage crawling, data cleaning, and talent mapping, standardizing a repetitive sourcing process into a reusable internal research pipeline.

PROJECT EXPERIENCE

MappingClaw: AI Sourcing Automation Workflow

Jan – Mar 2026

Workflow Designer and Tool Builder, OpenClaw for MiraclePlus

- Developed a department-level AI sourcing pipeline to address the high manual cost of identifying AI researchers, automating lead discovery, homepage crawling, paper/profile extraction, customized outreach drafting, CSV export, and Feishu-based delivery.
- Architected the workflow around Bright Data MCP search, Mapping-Skill field extraction, GLM-4 text generation, Tencent Cloud deployment, and Feishu Bot notifications, enabling the research team to review standardized outputs in a shared workspace.
- Defined a researcher data schema covering name, institution, research direction, representative publications, contact indicators, and relevance tags; implemented batch deduplication and cleaning rules to improve downstream interview screening.
- Operationalized the workflow as a reusable internal research tool, reducing outsourced data-collection cost from approximately RMB 20,000/month to about RMB 400/month in server and token costs.

Campus AI Learning Assistant Agent

2025 – 2026

Product Lead, Prompt Engineering and Iteration Lead

- Led early-stage product definition for a campus AI learning assistant by collecting 300+ student survey responses and evaluating 8 candidate scenarios through a "technical reliability x usage frequency" framework.
- Prioritized course Q&A, coding assistance, and review-material generation, while explicitly excluding paper-writing and low-frequency schedule-management features from the first release to reduce hallucination and scope risk.

- Drafted the product requirements document and designed domain-specific system prompts for mathematics, programming, and humanities tasks, improving response accuracy by 35% on the internal test set.
- Analyzed 47 closed-beta feedback items and identified multi-turn memory as the primary usability bottleneck; redesigned the context-memory mechanism and reduced related complaints to zero in the subsequent testing round.
- Coordinated 3 product iterations, increasing formula-recognition accuracy from 72% to 91% and validating sustained usage of the error-log synchronization feature for review and mistake tracking.

RAGShopGuide: Retrieval-Augmented Product Consultation System

May 2026

RAG Backend and Mobile Interface Developer; Code: github.com/Arreb-01/ragshopguide

- Developed a retrieval-augmented product consultation system that integrates structured product knowledge, semantic retrieval, and large-model generation to provide source-grounded shopping recommendations and product comparisons.
- Constructed the backend architecture with FastAPI, Chroma, and Doubao API, covering product-data ingestion, multi-source knowledge chunking, vector retrieval, prompt orchestration, and streaming response delivery.
- Designed a structured response protocol to preserve source attribution, product references, and comparison relationships; implemented retrieval-based fallback logic to improve consistency when model outputs omitted required evidence.
- Implemented Kotlin and Jetpack Compose interface components for conversational responses, recommended-question prompts, source evidence blocks, product cards, and comparison views, linking RAG reasoning to a mobile advisory experience.

Circuit Hero: Interactive Circuit Learning Game

Mar – May 2026

Core Interaction and Learning Logic Developer; Code: github.com/Arreb-01/circuit-hero

- Developed a six-chapter web-based educational game for students aged 12–16, translating closed circuits, switches, series circuits, parallel circuits, mixed circuits, and applied wiring tasks into scaffolded learning challenges.
- Implemented the workspace interaction layer with drag-and-drop components, port-to-port wiring, undo/redo, switch toggling, contextual hints, reset actions, particle-style current feedback, star ratings, and progress saving.
- Constructed a BFS-based circuit verification module by modeling components, ports, and wires as graph structures, enabling the system to detect valid circuits, missing loops, short circuits, and incorrect series/parallel topology.
- Developed the multi-page front-end flow and a Node.js/Express/SQLite backend for registration, login, and level-progress persistence; added Playwright E2E tests to verify core learning and account flows.

Lijin Pattern Memory: Interactive Web Platform for Li Ethnic Patterns

2025

Prototype Designer, Front-end Developer, and AI Generation Pipeline Member

- Designed and developed an interactive web platform integrating cultural education, pattern-asset browsing, and AI-assisted generation to broaden public engagement with Li ethnic motifs and support low-threshold creative reuse.
- Led Figma high-fidelity prototyping for an "AI-assisted design" workflow, balancing generation quality, operational simplicity, pattern interpretation, and visual exploration for non-specialist users.
- Curated and cleaned 500+ Li brocade pattern samples, fine-tuned Stable Diffusion for motif-consistent generation, and optimized API/front-end rendering to reduce average generation latency from 30s to 8s.
- Implemented the front end with HTML/CSS/JavaScript; the project reached 1,000+ exhibition visitors and 92% user satisfaction, and won the Design Creativity Award at the Hainan International Designers Competition.

AWARDS & HONORS

- Third Prize, Guangdong-Hong Kong-Macao Greater Bay Area Financial Mathematics Modeling Competition, Dec 2025.
- Third Prize, National College Student Digital Media Technology Works and Creativity Competition, Sep 2025.
- Design Creativity Award, Hainan International Designers Competition, Sep 2025.
- Second Prize, China Collegiate Computer Design Competition, Hainan Division, Jun 2025.
- National First-Class Athlete in Fencing.

LEADERSHIP & SERVICE

Li'an Coastal School

Oct 2024 – Apr 2025

Volunteer AI and Programming Teacher, International School Volunteer Service Team

- Delivered weekly AI and programming lessons to fourth-grade students, translating abstract computational concepts into age-appropriate examples, visual demonstrations, and interactive exercises.

TECHNICAL SKILLS

- **Programming & Development:** Python, JavaScript, HTML/CSS, Java, Kotlin, SQL; FastAPI, Node.js, Express, SQLite, REST APIs, SSE streaming APIs
- **AI, RAG & Data Analysis:** RAG pipeline design, Chroma vector retrieval, prompt engineering, structured output protocols, Doubao API, GLM-4, data cleaning, feature engineering, Python-based statistical analysis
- **Product & Research Methods:** product requirements documentation, scenario prioritization, user survey analysis, closed-beta feedback analysis, PANAS-based affect measurement, interaction logging, rule-based evaluation
- **Automation & Creative Computing:** OpenClaw workflow design, Bright Data MCP, Feishu Bot integration, Tencent Cloud deployment, Stable Diffusion fine-tuning, Figma prototyping, audiovisual interaction systems, Playwright E2E testing